

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming (Class Test)

Course Name	Python Programming
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Topic	CO3-AT2-CLASS TEST
Question	Q3 — Dictionary Operations on Employee Data (1 Mark)

Q3. [1 Mark]

Given `emp = {'id': 101, 'name': 'Ravi', 'dept': 'CSE'}`, write Python code to:

- (a) Add key 'salary' with value 50000.
- (b) Update department to 'AI'.
- (c) Display all keys, values, and items.

Approach / Design

- **Adding a key:** `emp['salary'] = 50000` inserts a new key-value pair into the dictionary.
- **Updating a value:** `emp['dept'] = 'AI'` overwrites the existing value for the 'dept' key.
- **Displaying:** `emp.keys()`, `emp.values()`, and `emp.items()` return view objects for the dictionary's keys, values, and key-value pairs respectively; wrapping each in `list()` makes them printable.

Python Program

```
"""
CO3-AT2 - Class Test
Q3: Dictionary Operations on Employee Data [1 Mark]
-----
Given emp = {'id': 101, 'name': 'Ravi', 'dept': 'CSE'}:
(a) Add key 'salary' with value 50000.
(b) Update department to 'AI'.
(c) Display all keys, values, and items.
"""

emp = {'id': 101, 'name': 'Ravi', 'dept': 'CSE'}

# (a) Add key 'salary' with value 50000
emp['salary'] = 50000

# (b) Update department to 'AI'
emp['dept'] = 'AI'

# (c) Display all keys, values, and items
print("Keys:", list(emp.keys()))
print("Values:", list(emp.values()))
print("Items:", list(emp.items()))
```

Sample Output

```
Keys: ['id', 'name', 'dept', 'salary']
Values: [101, 'Ravi', 'AI', 50000]
Items: [('id', 101), ('name', 'Ravi'), ('dept', 'AI'), ('salary', 50000)]
```

Conclusion

Starting from `emp = {'id': 101, 'name': 'Ravi', 'dept': 'CSE'}`, assigning `emp['salary']` adds a new key-value pair, and reassigning `emp['dept']` updates the existing department value in place. The final dictionary `{'id': 101, 'name': 'Ravi', 'dept': 'AI', 'salary': 50000}` is then inspected using `keys()`, `values()`, and `items()`, confirming that both the addition and the update were applied correctly.